

Axon initial segment dynamics during associative fear learning

Received: 1 June 2023

Accepted: 23 October 2025

Published online: 23 December 2025



Chloé Maëlle Benoit ^{1,2,9}, Dan Alin Ganea ^{1,2,9} ✉, Ricardo Paricio-Montesinos ¹, Julia Esser ¹, Christian Thome ^{3,4}, Jan Maximilian Janssen ^{3,4}, Andrea Sattin ⁵, Sabrina Milena Innocenti², Sabine Krabbe ¹, Andreas Lüthi ^{6,7}, Tommaso Fellin ⁵, Maren Engelhardt^{3,4} & Jan Gründemann ^{1,2,8} ✉

The axon initial segment (AIS) is the site of action potential generation and exhibits structural and functional plasticity upon adaptation of neuronal excitability and disease. Performing *in vivo* longitudinal two-photon imaging of AIS in the medial prefrontal cortex of male mice, we reveal dynamic AIS remodeling during associative fear learning and extinction. These results demonstrate that AIS plasticity is not only crucial for homeostatic adaptation but also a hallmark of memory formation.

Learning and memory are fundamental to adapt behavioral strategies in new environmental circumstances. Behavioral changes are supported by functional plasticity at the neuronal circuit level, which correlate with neuronal morphology changes^{1–3}. Hence, revealing the mechanisms of experience-dependent structural plasticity in neuronal subtypes and pathways *in vivo* is crucial to understand the principles that link adaptive behaviors with cellular and neural circuit mechanisms of memory formation^{4,5}.

The axon initial segment (AIS) is a specialized neuronal compartment responsible for action potential initiation, which is supported by high ion channel density^{6,7} rendering the AIS the final site of synaptic integration⁸. In addition, the AIS plays a crucial role in homeostatic maintenance of neuronal excitability and network function during sensory processing and learning^{9–11}.

Ex vivo and *in vitro* studies identified the AIS as a site of neuronal plasticity. Changes in AIS location and length correlate with changes in neuronal excitability^{10,12,13}. In contrast to synapse-specific weight adjustments, AIS plasticity rescales the global input–output function of a neuron, providing a fundamental, computationally distinct, synapse-independent plasticity mechanism during development^{14,15}, aging¹⁶, sensory stimulation or deprivation^{11,17–19} and disease states^{20–26}. In addition to homeostatic adjustments, AIS remodeling could be crucial for plasticity during learning and memory or the recruitment of individual neurons to subnetworks of shared excitability as well as memory engrams, given that AIS length adapts during NMDA

receptor-mediated plasticity²⁷. However, it remains unclear if structural AIS plasticity occurs during associative learning and memory formation *in vivo* due to the lack of AIS-specific longitudinal imaging techniques. We hypothesized that changes in AIS length act as a cellular mechanism of associative learning in medial prefrontal cortex (mPFC), a brain area involved in adaptive threat behavior and memory formation^{28–32}. Using a new longitudinal live imaging approach designed to monitor AIS *in vivo*, we aimed to test if AIS of mPFC neurons are plastic during fear learning and extinction.

Tracking AIS length *in vivo* across fear learning

To visualize AIS morphological changes *in vivo*, we used a transgenic mouse line expressing a cre-dependent Ankyrin-G-GFP (AnkG-GFP) fusion protein to label the AIS. AnkG is a scaffolding protein enriched at the AIS³³, thus allowing for detailed analysis of AIS geometry (for example, length) in the AnkG-GFP mouse line³⁴. AnkG-GFP mice were intercrossed with Cre-dependent tdTomato mice (Ai9) to visualize AIS and neuronal structures in mPFC excitatory neurons via viral expression of CaMKII-dependent Cre recombinase (Fig. 1a). *Ex vivo* immunofluorescence verified that the AIS label (GFP) colocalized with AnkG and Na_v1.6 channels across its entire length (Fig. 1b, stimulated emission depletion (STED) imaging to verify the overlap of AnkG and Na_v1.6 in the mPFC of wild-type mice; Extended Data Fig. 1). AnkG and sodium channel 1.6 (Na_v1.6) channels are both enriched at the AIS³⁵, verifying the specificity of the approach³⁴. To monitor the same AIS longitudinally *in vivo*

¹Deutsches Zentrum für Neurodegenerative Erkrankungen (DZNE), Bonn, Germany. ²Department of Biomedicine, University of Basel, Basel, Switzerland.

³Institute of Anatomy and Cell Biology, Johannes Kepler University, Linz, Austria. ⁴Clinical Research Institute for Neurosciences, Johannes Kepler University, Linz, Austria. ⁵Optical Approaches to Brain Function Laboratory, Istituto Italiano di Tecnologia, Genova, Italy. ⁶Friedrich Miescher Institute for Biomedical Research, Basel, Switzerland. ⁷University of Basel, Basel, Switzerland. ⁸Faculty of Medicine, University of Bonn, Bonn, Germany. ⁹These authors contributed equally: Chloé Maëlle Benoit, Dan Alin Ganea. ✉e-mail: dan.ganea4@gmail.com; jan.grundemann@dzne.de

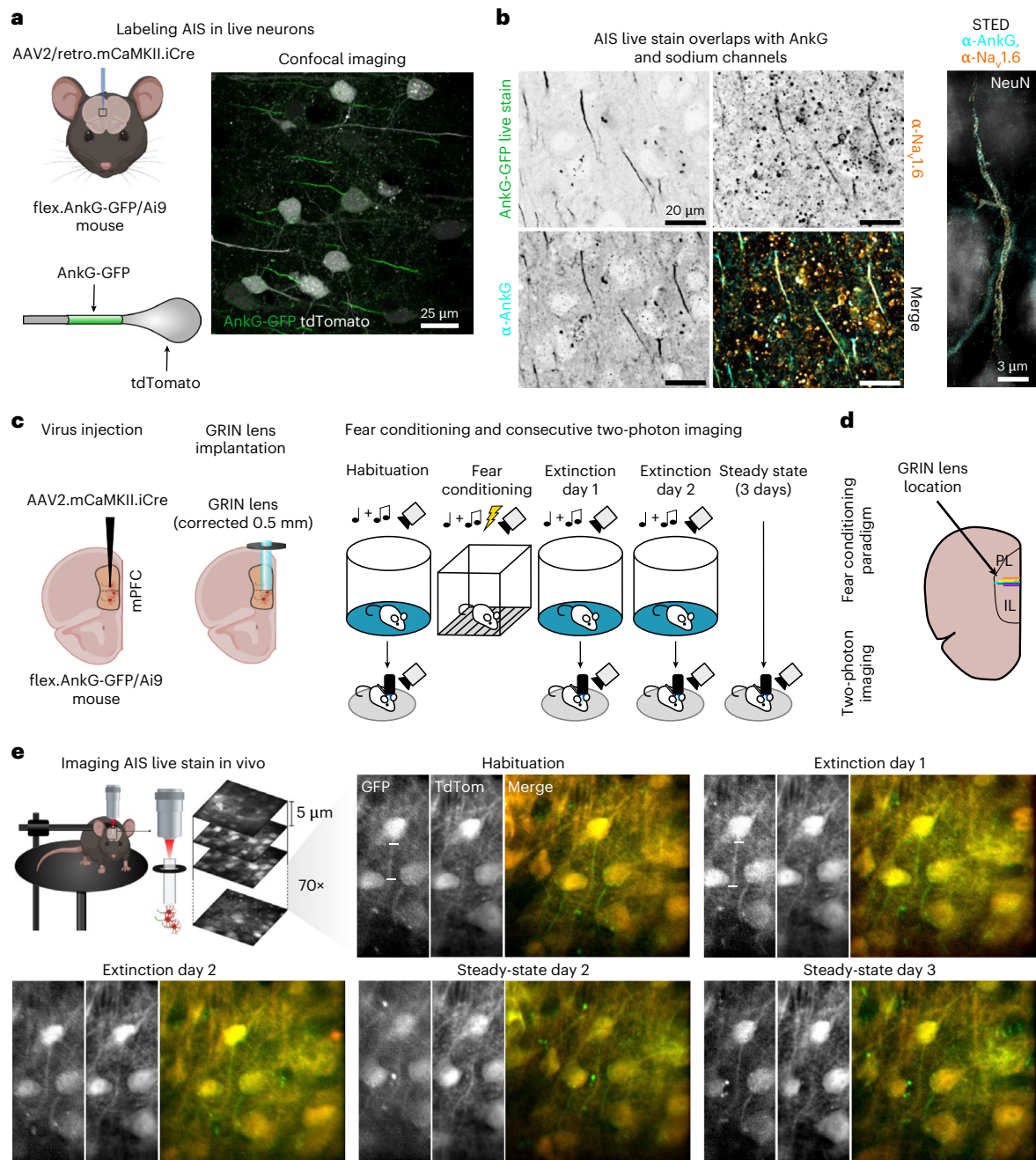


Fig. 1 | AIS in vivo imaging upon auditory fear conditioning. **a**, Injection of AAV2/retro.mCaMKII.iCre in the mPFC of flex.AnkG-GFP/Ai9 mouse (top left) inducing the expression of tdTomato (red) and GFP (green) respectively tagging the neuronal structure and the AIS (bottom left). High magnification ex vivo confocal image of the viral expression of the AnkG-GFP live label in a flex.AnkG-GFP/Ai9 mouse in mPFC. **b**, Immunofluorescence and confocal image of a flex.AnkG-GFP mouse mPFC expressing the intrinsic AIS GFP construct as well as antibody immunofluorescence staining against AnkG and Na_v1.6 (left). Immunofluorescence and STED super-resolution microscopy of antibody staining against AnkG and Na_v1.6 demonstrating signal colocalization of the two proteins in mPFC of wild-type mice (right). Short branches are optically cut AIS from neighboring neurons in this field of view. For details see Extended Data Fig. 1. **c**, Viral approach for tagging AIS in vivo and behavioral paradigm. Local stereotaxic injection of virus and aberration-corrected GRIN lens implantation

in mPFC followed by a behavioral paradigm and two-photon imaging sessions (4 days of freely moving auditory fear conditioning and 3 days of steady state). **d**, GRIN lens distal end (lens fronts of individual animals represented as horizontal lines) in mPFC for all mice ($N = 6$). PL, prelimbic cortex; IL, infralimbic cortex. **e**, Schematic of GRIN-lens-based two-photon imaging approach in mPFC allowing the detection of AIS across days. Example of in vivo two-photon images of mPFC neurons expressing AnkG-GFP and tdTomato across several days of the paradigm (habituation, extinction days 1 and 2, and steady-state days 2 and 3; similar results observed in $N = 6$ mice). White lines indicate start and end of AIS for length measurement. Mouse cartoon in **a** created in BioRender. Benoit, C. (2025) <https://BioRender.com/yzwn720> and in **e** partially created in BioRender. Benoit, C. (2025) <https://BioRender.com/clhnp1f>, respectively. Brain slice in **c** partially created in BioRender. Benoit, C. (2025) <https://BioRender.com/l7ge0v8>.

during associative learning, we used two-photon imaging through an **aberration-corrected gradient refractive index (GRIN)-lens-based endoscope**^{36,37} together with auditory fear conditioning and extinction

as well as steady-state days where mice were not exposed to the behavioral paradigm or any changes as a control for plasticity effects (Fig. 1c and Extended Data Fig. 2).

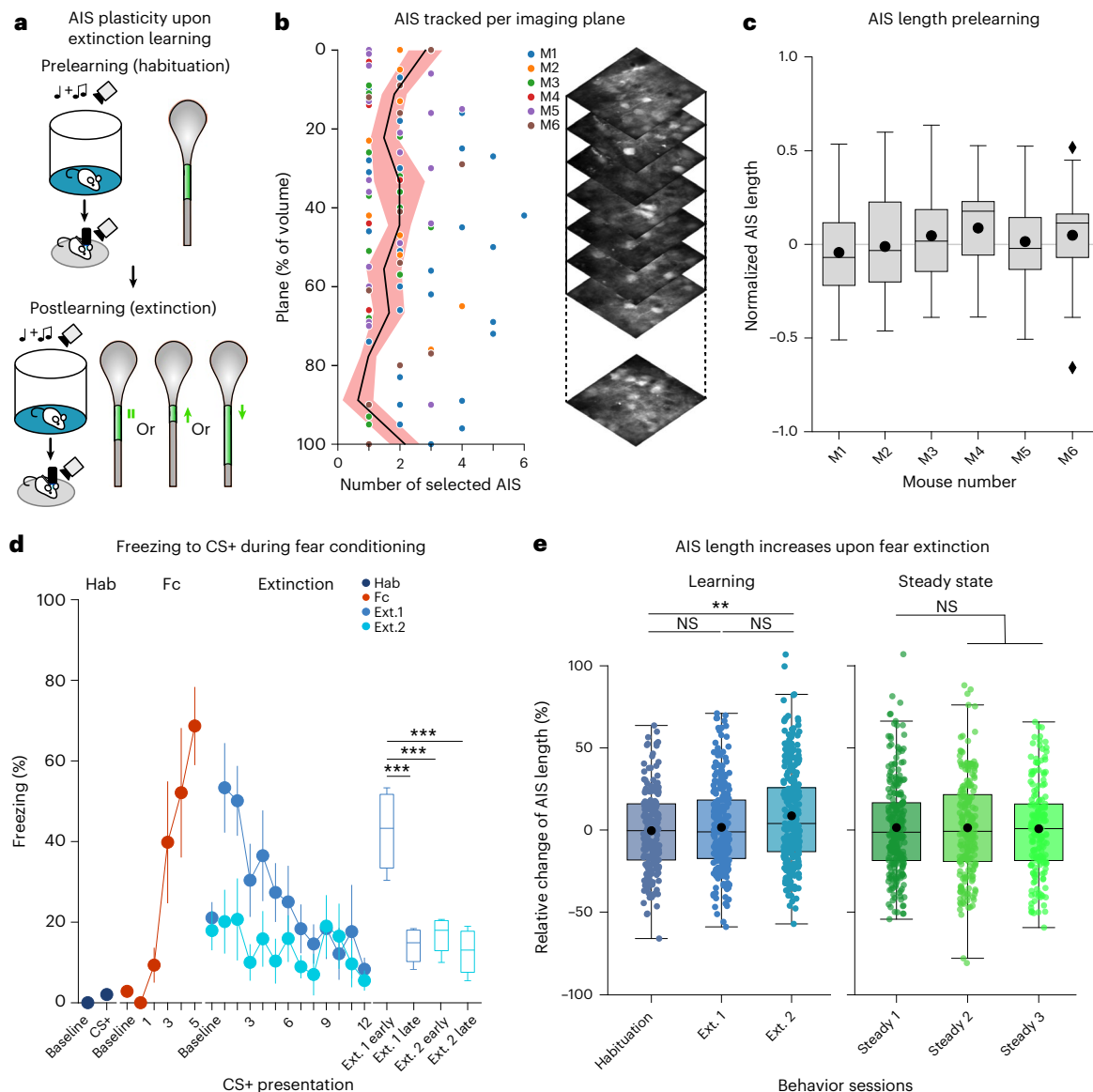


Fig. 2 | mPFC AIS exhibit plasticity following extinction learning. **a**, Schematic of potential AIS plasticity taking place following extinction learning. **b**, Overall number of AIS tracked per imaging plane for all mice (2.06 ± 0.12 AIS selected per plane) during the first day of imaging. Black and red lines represent mean $\pm$ s.e.m. **c**, Length of AIS prelearning for all mice (-0.002 ± 0.017 normalized AIS length during habituation (equation (1)). All pairs are not significant, Kruskal–Wallis test with Dunn’s post hoc correction, $n = 212$ AIS from $N = 6$ mice). **d**, Behavioral analysis of the animals’ freezing behavior (Ext. 1 early versus Ext. 1 late: $P = 1.04 \times 10^{-4}$, Ext. 1 early versus Ext. 2 early: $P = 4.59 \times 10^{-4}$, Ext. 1 early versus

Ext. 2 late: $P = 2.83 \times 10^{-5}$, Kruskal–Wallis test ($P = 4.47 \times 10^{-5}$) with Dunn’s post hoc correction, $N = 6$ mice). **e**, Relative change in AIS length in relation to the population average prelearning (%) (Hab versus Ext. 2: $P = 0.0072$, Kruskal–Wallis test ($P = 0.0034$) with Dunn’s post hoc correction, total number of AIS = 1605 out of $N = 6$ mice). Box plots represent median, 2nd quartile, 3rd quartile, minimum and maximum. Dots indicate means and diamonds are outliers. * P and ** P values smaller than 0.01 and 0.001, respectively. Fc, fear conditioning; Ext. 1, extinction day 1; Ext. 2, extinction day 2; Hab, habituation; NS, non-significant.

Mice underwent a four-day fear conditioning and extinction paradigm in which they learned to associate a mild foot shock (unconditioned stimulus, US) with an auditory cue (conditioned stimulus, CS+). During fear extinction, animals were reexposed to repeated CS+ resulting in extinction of conditioned freezing behavior. Each freely moving fear conditioning and extinction session was followed by head-fixed two-photon imaging to visualize and track AIS length of mPFC neurons, while the final 3 days of steady-state two-photon imaging were devoid of any cues or contextual elements to test for potential learning-unrelated changes in AIS length (Fig. 1c). Two-photon volume scans were performed in mPFC (Fig. 1d, predominantly the **infralimbic subdivision**, $N = 6$ mice) and individual AIS were longitudinally visualized across the learning paradigm (Fig. 1e).

Prefrontal cortex AIS exhibit plasticity following extinction learning

Next, we measured AIS length in vivo (Methods) across habituation and extinction days 1 and 2 to test how AIS length changes with learning (Fig. 2a). AIS were sampled from different imaging planes and depth similarly for all mice (Fig. 2b). Before learning during the habituation day, AIS length did not differ between mice (number of AIS: 2.06 ± 0.12 AIS per imaging plane per mouse, Fig. 2c and Extended Data Fig. 3; mean normalized AIS length change: -0.002 ± 0.017 , Fig. 2c, Methods and equation (1)). Upon fear conditioning, animals acquired a freezing response to the CS+ (CS+ freezing: $68.7 \pm 9.7\%$, CS– freezing: $30.1 \pm 12.2\%$, $N = 6$ mice, Fig. 2d and Extended Data Fig. 1c). Upon CS+ reexposure during the first fear extinction session, mice showed

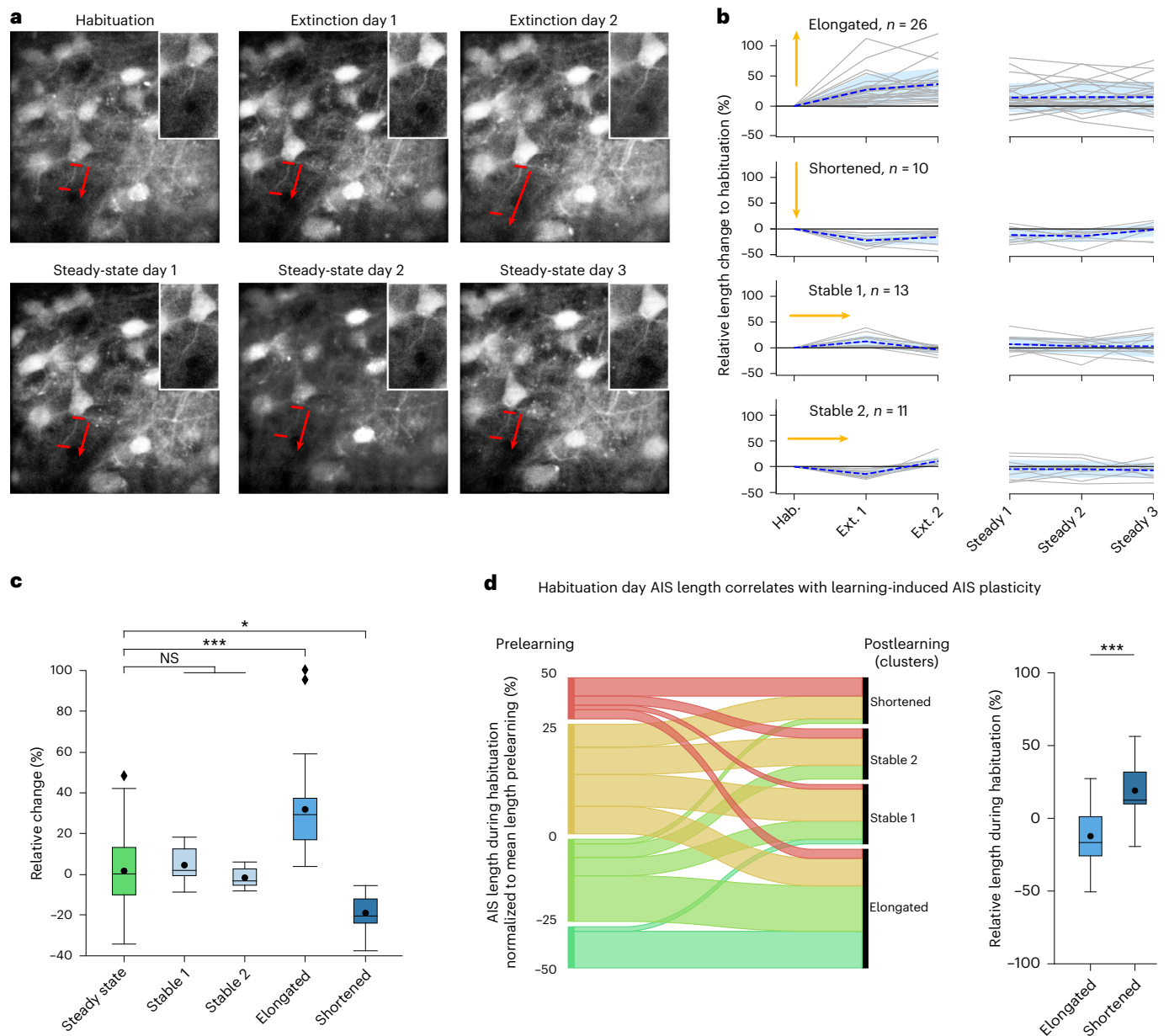


Fig. 3 | Evolution of individual AIS length changes during fear extinction learning. **a**, Example of in vivo tracking of one example AIS across learning and steady-state days (example neuron from elongated cluster). **b**, K-means clusters of AIS length dynamics across imaging days ($n = 60$ AIS from $N = 6$ mice). **c**, Change in relative AIS length within learning per cluster compared to the steady-state days (Steady state versus Stable 1: $P = 0.99$, Steady state versus Stable 2: $P = 0.99$, Steady state versus Elongated: $P = 2.32 \times 10^{-7}$, Steady state versus Shortened: $P = 0.011$, Kruskal–Wallis test ($P = 7.28 \times 10^{-11}$) with Dunn’s

post hoc correction, $n = 60$ AIS from $N = 6$ mice). **d**, Sankey plot of the distribution of AIS across the different clusters according to their normalized length prelearning (equation (1)). AIS length during habituation normalized to the population mean in percent ($P = 7.44 \times 10^{-4}$, two-sided Mann–Whitney U test) for the Elongated and Shortened clusters ($n = 36$ AIS from $N = 6$ mice). Box plots represent median, 2nd quartile, 3rd quartile, minimum and maximum. Dots indicate mean and diamonds outliers. * P and *** P values smaller than 0.05 and 0.001, respectively.

initially high CS+ selective freezing levels indicating fear memory consolidation (Extinction 1 early CS+ freezing: $42.6 \pm 2.5\%$ versus CS–freezing: $10.2 \pm 1.3\%$, $P = 1.12 \times 10^{-5}$, Mann–Whitney U test, $N = 6$ mice, Extended Data Fig. 1b), which extinguished during consecutive CS+ presentations and consolidated overnight (Extinction 1 early versus Extinction 1 late: $P = 1.04 \times 10^{-4}$, Extinction 1 early versus Extinction 2 early: $P = 4.59 \times 10^{-4}$, Extinction 1 early versus Extinction 2 late: $P = 2.83 \times 10^{-5}$, Kruskal–Wallis test ($P = 4.47 \times 10^{-5}$) with Dunn’s post hoc correction, Fig. 2d). After each behavioral training session (except fear conditioning), AIS were visualized under a two-photon microscope to relate behavioral changes with AIS length changes. At the population level, AIS significantly elongated following the second

day of fear extinction learning relative to habituation day (Habituation versus Extinction 1, $P = 0.938$, Habituation versus Extinction 2, $P = 0.0072$, Extinction 1 versus Extinction 2, $P = 0.024$, Kruskal–Wallis test ($P = 0.0034$) test with Dunn’s post hoc correction, Fig. 2e and Extended Data Fig. 4). During subsequent nonlearning, steady-state imaging days, where animals were not exposed to any learning session, average AIS length returned to its original size and remained steady throughout subsequent imaging days (Habituation day before fear conditioning versus Steady-state days 1, 2 and 3: $P = 0.978$, Kruskal–Wallis test, Fig. 2e and Extended Data Fig. 4). Taken together, this indicates that following fear extinction learning, the average AIS length in mPFC increases, while AIS are stable outside of the conditioning

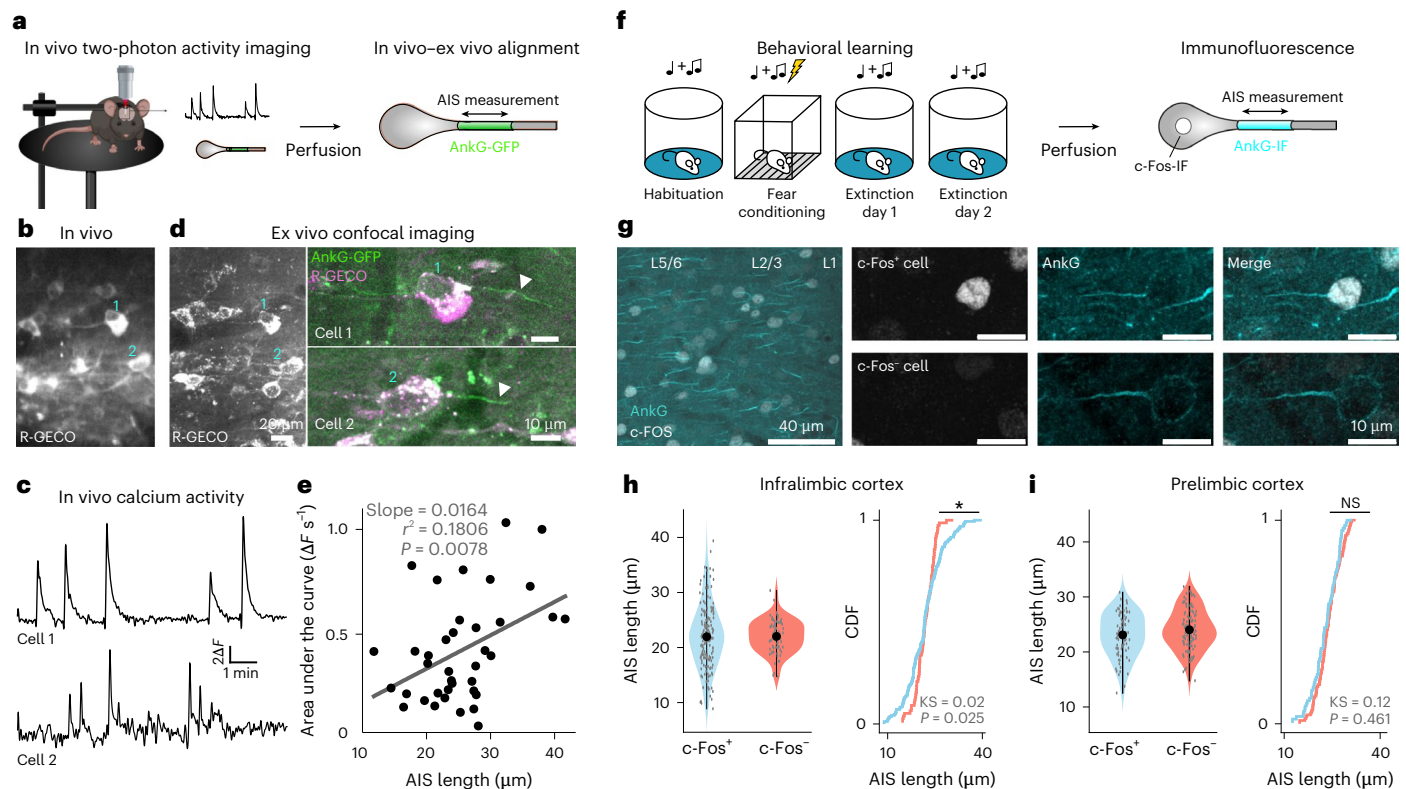


Fig. 4 | AIS dynamics relate to neuronal excitability and c-Fos expression following fear extinction learning. **a**, Experimental approach to link in vivo neuronal activity to AIS length ex vivo. **b**, R-GECO expression in vivo (similar results observed in $n = 3$ mice). **c**, Example Ca^{2+} activity traces of cells in **b**. **d**, Confocal ex vivo overview image of the cells from **b**. Arrowheads point to Ankg-GFP-positive AIS of the two example cells from **b** (similar results observed in $n = 3$ mice). **e**, Correlation of mean area under the curve of in vivo calcium events and ex vivo measure AIS length after fear extinction (two-sided Pearson correlation, $n = 38$ aligned neurons from $n = 3$ mice). **f**, Experimental approach to link AIS length to memory engrams using immunofluorescence (IF) staining of c-Fos and Ankg. **g**, c-Fos (white) and Ankg (cyan) expression patterns in mPFC after fear extinction (left). Example c-Fos⁺ and c-Fos⁻ neurons and their respective

AIS (right). **h**, Violin plots of AIS length for c-Fos⁺ and c-Fos⁻ cells in infralimbic cortex (left, c-Fos⁺ mean = $21.965 \pm 0.439 \mu\text{m}$, c-Fos⁻ mean = $22.018 \pm 0.338 \mu\text{m}$). Population cumulative distribution function (CDF, right, $P = 0.025$ and CDF statistic = 0.20, $n = 245$ AIS, $N = 9$ mice; two-sided Kolmogorov–Smirnov test). **i**, Violin plots of AIS length for c-Fos⁺ and c-Fos⁻ cells in prelimbic cortex (left, c-Fos⁺ mean = $23.110 \pm 0.451 \mu\text{m}$, c-Fos⁻ mean = $24.058 \pm 0.354 \mu\text{m}$). Population CDF (right, $P = 0.461$ and CDF statistic = 0.12, $n = 186$ AIS, $N = 8$ mice; two-sided Kolmogorov–Smirnov test). Violin plots represent 2nd quartile, 3rd quartile, minimum and maximum. Dots indicate mean. * P values smaller than 0.05. KS, Kolmogorov–Smirnov. Mouse cartoon in **a** was partially created in BioRender. Benoit, C. (2025) <https://BioRender.com/kth75iv>.

paradigm, suggesting that structural changes at the AIS are a cellular correlate of learning-related plasticity in mPFC.

Diverse plasticity patterns of mPFC AIS length upon fear extinction learning

To investigate the dynamics of individual AIS across fear learning and extinction, we subsampled AIS that could be longitudinally tracked throughout all imaging sessions (60 out of an average of 268 daily imaged AIS, 22%) and measured their length relative to the habituation session (Fig. 3a). *K*-means clustering revealed four distinct clusters of dynamic changes of AIS length (Fig. 3b). The largest group (43.3%), termed elongated cluster, increased in length ($36.4 \pm 5.3\%$ relative length change to habituation after Extinction 2) following fear extinction learning, which was in line with the general population average. However, opposed to the average population dynamics (Fig. 2), AIS length did not return to baseline after extinction consolidation but maintained its elongated state (Elongated cluster Habituation versus Steady-state day 1: $P = 3.24 \times 10^{-5}$, Habituation versus Steady-state day 2: $P = 2.56 \times 10^{-4}$, Habituation versus Steady-state day 3: $P = 6.21 \times 10^{-5}$, Kruskal–Wallis test ($P = 2.04 \times 10^{-6}$) with Dunn's post hoc correction, Fig. 3b).

A second, smaller cluster contained AIS (16.7%, shortened cluster) that exhibited a decrease in length ($-15.8 \pm 4.3\%$) following fear extinction. This decrease was less pronounced relative to the change

observed for the elongated cluster and AIS tended to return to their original length during steady-state sessions. Additionally, we observed two stable clusters. These clusters exhibited initially slight decreases (stable cluster 1: relative change: $-3.6 \pm 2.0\%$, 21.6% of AIS) or increases in length (stable cluster 2: relative change: $10.9 \pm 2.6\%$, 18.3% of AIS) after conditioning. However, these AIS reverted to their original length and maintained it throughout subsequent sessions. The across-day length change during learning for stable clusters was not significantly different to the total change across days during steady-state sessions, indicating that the length change observed for stable clusters was not different to changes in length that AIS undergo during nonlearning conditions. However, the change in length was significantly different between steady-state days and elongated or shortened plastic clusters (Steady state versus Stable 1: $P = 0.99$, Steady state versus Stable 2: $P = 0.99$, Steady state versus Elongated: $P = 2.23 \times 10^{-7}$, Steady state versus Shortened: $P = 0.011$, Kruskal–Wallis test ($P = 7.28 \times 10^{-11}$), Dunn's post hoc correction, Fig. 3c). Overall, we observed diverse plasticity of AIS length following fear extinction learning (Extended Data Fig. 5). AIS of mPFC excitatory neurons exhibited significant learning-dependent length changes relative to preconditioning states. The elongated cluster contained both the highest AIS number and maintained an increase in length throughout all imaging days. Finally, tracking AIS plasticity cluster composition by initial length revealed that elongated and shortened AIS clusters originate from the pool of shortest or longest AIS,

respectively (Fig. 3d, $P = 7.44 \times 10^{-4}$, Mann–Whitney U test), suggesting that AIS length changes occur within a predetermined physiological dynamic range in mPFC, beyond which elongation or shortening does not result in additional changes to neuronal excitability^{38,39}.

AIS length correlates with calcium dynamics

To test if AIS length is linked to neuronal activity, we combined two-photon calcium imaging of mPFC excitatory neurons with post hoc AIS length measurements. Mice were exposed to the fear learning and extinction paradigm as described. Following the second day of extinction, spontaneous neuronal activity of mPFC neurons was measured with the red-shifted calcium indicator R-GECO and AIS length measured subsequently in horizontal brain slices ex vivo (Fig. 4a–d). Thirty-eight neurons from three mice could be matched between in vivo two-photon imaging and post hoc ex vivo fluorescence microscopy. Interestingly, in vivo calcium event rates of individual neurons did not significantly correlate with AIS length (Extended Data Fig. 6). However, AIS length positively correlated with the area under the curve of individual calcium events (Fig. 4e) indicating that increased AIS length is associated with bursts of high frequency action potentials in mPFC neurons.

Fear extinction engrams are associated with long and short AIS

Finally, we tested if AIS length is associated with memory allocation. Mice were exposed to the fear conditioning and extinction paradigm (Fig. 4f and Extended Data Fig. 7a) and subsequently perfused. This approach prevented us following individual AIS in vivo, yet provided access to large numbers of neurons. Strikingly, we found that AIS of c-Fos⁺ neurons (Fig. 4g) exhibited a large distribution of AIS lengths, including neurons with very short and very long AIS in infralimbic cortex after fear extinction (Fig. 4h). In contrast, this difference could not be observed in prelimbic cortex, a brain area thought to be involved in fear recall⁴⁰ (Fig. 4i and Extended Data Fig. 7b,c). These findings suggest that dynamic changes in infralimbic cortex AIS length upon fear extinction occur in c-Fos-activated ensembles, which are enriched with shortened and lengthened AIS. These changes in AIS length will allow for neuronal output adjustments during engram formation upon extinction memory recall, a function that was previously associated with infralimbic cortex³⁹. Additionally, this effect is brain area- and learning-specific, given that differences in AIS lengths of c-Fos⁺ neurons could not be observed in prelimbic cortex after extinction learning (see above), yet emerged after fear recall training (Extended Data Fig. 7). This supports the notion that AIS length is dynamic upon learning and plasticity-inducing events in an engram- and brain area-specific fashion.

mPFC pyramidal neurons have previously been identified as a site of functional plasticity during extinction learning⁴¹. Using an AIS marker line and GRIN-lens-based two-photon imaging, we visualized and tracked AIS length longitudinally across a 4-day fear conditioning and extinction paradigm. We identified population level changes in AIS length upon fear extinction learning. Overall, increased AIS length was associated with fear extinction learning. Nevertheless, at the single cell level, distinct clusters of mPFC neurons with elongated or shortened AIS emerged upon extinction learning, suggesting that AIS length changes could map onto distinct populations of mPFC pyramidal neurons^{42–45} recruiting them into opposing functional ensembles with increased or decreased excitability. Therefore, bidirectional AIS plasticity could be a mechanism to counterbalance the weight between distinct mPFC subnetworks, for example, neurons of functionally distinct output pathways or opposing engrams for the formation and extinction of associative memories^{41,44,46}.

Indeed, neurons of the elongated cluster did not revert their AIS length post-extinction learning, suggesting that persistent AIS elongation of this cluster is associated with elevated excitability and supports long term extinction memory in mPFC networks^{41,47–49}.

Interestingly, before associative learning, AIS recruited into the elongated cluster tend to be shorter while those recruited to the shortened cluster tend to be longer, suggesting that AIS length changes occur most dynamically at the extreme ends of the length distribution within a defined dynamic range and structural limitation, where they might exert their strongest impact on action potential initiation and excitability^{38,39}. Prelearning AIS length might thus be a hallmark of availability for neuronal plasticity of mPFC pyramidal neurons and their recruitment to memory engrams. This is further supported by our finding that infralimbic c-Fos⁺ neurons exhibit a broader AIS length distribution than c-Fos[−] neurons following fear extinction learning, suggesting that fear extinction engrams are linked to both, elongation and shortening of AIS clusters. AIS length-dependent up- or downregulation of neuronal excitability within c-Fos⁺ engrams would be crucial to mediate distinct mPFC-output pathways that drive learning-specific behaviors. In contrast, c-Fos[−] neurons are typically not considered part of an engram and exhibited a narrower AIS length distribution after learning. This could result from both, neurons that did not change their AIS length and neurons that adjusted their AIS length toward the mean of the c-Fos[−] population, which might be a potential homeostatic mechanism balancing activity and dampening output of the non-engram network. Together, our findings support the notion that learning is dependent on bidirectional control of AIS length and in turn adaptive neuronal activity within the memory engram to balance the activity of neuronal types with distinct functions.

Here we have shown that dynamic changes in AIS length occur during associative learning and extinction in vivo and might be a cellular mechanism of plasticity regulation during memory formation. Together with recent ex vivo studies, this suggests that AIS length changes might be a general mechanism for plasticity regulation and memory formation in other brain areas and behaviors beyond mPFC and fear learning, respectively^{11,27,50,51}. Furthermore, AIS dynamics could contribute to both the formation of specific cue-driven changes in the input–output function of tuned neurons as well as the regulation of excitability in behavior-specific neuronal ensembles or memory engrams driving behavior and memory recall in a state-dependent fashion^{52–55}. Given that the AIS is perturbed in a number of neurological diseases^{21,23–26}, including neurodegenerative conditions like Alzheimer's disease⁵⁶, maladaptation of AIS plasticity might play an important role in disorders of memory loss, cognitive decline and behavioral changes long before neuronal cell loss occurs.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41593-025-02152-5>.

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Methods

Animal subjects

Experiments were performed in accordance with the institutional guidelines of the University of Basel (Tierschutz) and the German Center for Neurodegenerative Disorders, and were approved by the Cantonal Veterinary Office of Basel-Stadt and the Landesamt für Natur-, Umwelt- und Verbraucherschutz NRW. Experiments were conducted in 9–15-week-old male C57Bl/6J, AnkG-GFP or AnkG-GFP::Ai9 mice. Cre-dependent AnkG-GFP mice (B6;129SV-ank3^{tm1DCI/HD}) were provided by Maren Engelhardt, Johannes Kepler University Linz³⁴. AnkG-GFP mice were crossed with Cre-dependent tdTomato reporter (Ai9) mice (Gt(ROSA)26Sor^{tm9(CAG-tdTomato)Hze}, JAX: IMSR JAX:007909 (ref. 57)) to generate the AnkG-GFP::Ai9 mice. Mice were housed on a 12-hour light/dark cycle and had ad libitum access to food and water.

Surgical procedures

Systemic analgesia (0.1 mg kg⁻¹ buprenorphine, subcutaneous (s.c.)) was provided 30 min before the start of the surgery. Mice were anesthetized with isoflurane (4% induction, 1.5–2.0% maintenance, Henry Schein) in oxygen-enriched air (95%, Oxymat III, Weinmann). Stereotactic surgeries were performed with a high-precision stereotaxic apparatus (Model 1900, Kopf Instruments). Body temperature was maintained at ca. 36 °C (Rodent warmer, 53800 M, Stoelting) and eyes were covered with a protective cream (Bepanthen, Bayer). Local analgesia was provided s.c. above the skull before incision of the skin (Lidocain (10 mg per kg) and Ropivacain (Naropin, 3 mg per kg)). Viral vectors (250 nl AAV2/retro.mCaMKII.iCre.WPRE-hGFP(A), VVF Zurich and 250 nl, AAV1.Syn.NES-jRGECO1a.WPRE.SV40, Addgene 100854-AAV1 for in vivo recordings of neuronal activity) were injected at coordinates AP 1.8, ML –0.35, DV –2.0 mm with the help of a hydraulic micropositioner (Model 2650, Kopf) with a pulled glass pipette (2-000-0001, Drummond Scientific) and a pressure ejection system (Picospritzer, Parker). Meloxicam (Metacam, 5 mg per kg, s.c.) was injected as postoperative analgesia once a day for 3 consecutive days after surgery.

An aberration-corrected GRIN lens (diameter 0.5 mm, length 4.07 mm, focal length in water immersion 200 µm)^{36,37} was implanted during a second surgery (anesthesia and analgesia as previously described). A small craniotomy was drilled (Kyocera) above the virus injection coordinates, and a small track was cut with a 0.7 mm sterile needle (22 G, Terumo). The GRIN lens was then slowly advanced into the brain (DV –2.2 mm) and fixed to the skull with ultra violet light curable glue (Loctite 4305, Henkel). The skin was sealed with Scotchbond (3 M). A stainless-steel head bar (custom made) was attached to the skull to fix the animal during imaging sessions by using Vetbond (3 M) and dental acrylic (Paladur, Kulzer).

Behavioral paradigm

Behavioral experiments were performed during the animal's light period. Mice were acclimatized to the experimenter, two-photon scanning and head fixation for 3 days. Next, a 4-day auditory fear conditioning paradigm was performed in a habituation/test context (days 1, 3 and 4) and a fear conditioning context (day 2). During the habituation session (30 min), mice were presented with 5 intermingled CS+ and CS– (5 each, 12 kHz and 6 kHz, respectively) in a round plexiglass context (low light conditions, 1% acetic acid isoamyl ester). CSs were composed of 27 tone pips (200 ms, 75 dB) presented at a rate of 1.1 Hz (Tucker Davis Technologies, RZ6). Fear conditioning was performed in a 25 cm-square plexiglass box with a shock grid floor (Coulbourn, Noldus, bright light conditions, 70% ethanol smell, 0 min). Five CS+ were terminated by a 2 s, 0.65-mA foot shock immediately after the last tone pip and intermingled with five CS– presentations. During fear extinction learning sessions (days 3 and day 4, 35 min), mice were presented with 4 CS– and 12 CS+ in the habituation context. Immediately after each behavioral session, except for fear conditioning, mice were head fixed under a two-photon microscope for a structural

scan of AIS. Steady-state sessions (days 5–7) were conducted after the fear extinction days during which mice were imaged similarly to the habituation and learning days. Fear conditioning experiments were performed using Ethovision XT 14 (Noldus). Behavioral tracking was conducted using DeepLabCut⁵⁸. Freezing was detected automatically based on periods of absence of movement (threshold: 1 s) and checked manually for non-freezing episodes (for example, grooming) post hoc.

In vivo two-photon imaging

In vivo two-photon imaging was performed using a custom-built two-photon microscope (Independent NeuroScience Services). Mice were head fixed with a custom-designed head bar system and placed on a running wheel during imaging. The microscope was equipped with a resonant scanning system (CRS08K, Cambridge Technology) and a pulsed Ti:sapphire laser (Chameleon Vision S, Coherent) and controlled by ScanImage software (Version 5.7, Vidrio Technologies). We used a water dipping objective lens (N16XLWD-PF-16X CFI1WD, 0.80 NA, Nikon) to focus on the back end of the aberration-corrected GRIN endoscope^{36,37}. The light path included a dichroic mirror (T565lpxr, long pass, Chroma), barrier filters (green: ET510/80 m, red: ET630/75 m, Chroma) and photomultiplier tubes (PMT2101, Thorlabs). For both fluorescence channels, images were acquired at a resolution of 512 × 512 pixels, with 940 nm excitation and 30 frames per second. For calcium imaging experiments, the spontaneous activity of mPFC excitatory neurons was recorded via two-photon imaging of the calcium indicator R-GECO, over 10 min, following the second day of fear extinction learning. Images were acquired at a resolution of 512 × 512 pixels, with 1,020 nm excitation and 30 frames per second. All mice were habituated to the experimenter, head restraint and two-photon imaging in the same imaging setup for several days before the imaging experiments.

Image acquisition and processing

Corrected GRIN lenses were used to reduce effects of optical aberrations^{36,37}. Two-photon z-stack dual-color images (green or red) were acquired along a volume of ~400 µm (5-µm step, 250 frames per plane). After acquisition, the images were motion corrected using Matlab code adapted from Suite2P⁵⁹. The images of each channel were mean projected and merged to create dual-color images using ImageJ (Version 2.0). Signal intensities in both green and red channels were adjusted in ImageJ for visualization purpose by saturating the brightest 35% of the pixels per channel, per imaging plane to maintain an unbiased transformation between planes. Using Simple Neurite Tracer (SNT, ImageJ 2.0), AIS visible on the green channel but not red channel (Extended Data Fig. 8 shows examples of visual differences) were manually marked on each of the green channel images and a fitted path was applied to allow the measurement of AIS length. For GRIN-2P in vivo imaging, AIS start and end were visually defined by the experimenter as a sharp change of fluorescence in the in vivo 2P image. mPFC AIS typically start at the soma indicated by a sharp rise in fluorescence (Extended Data Fig. 9). For individual mice, in vivo imaged AIS were aligned manually across a number of days. For ex vivo confocal imaging, AIS intensity data was filtered using a Savitzky-Golay filter. The AIS location was defined as a rigid 80% drop from the peak fluorescence for confocal images. Analysis and measurement of the AIS (in pixels) were performed in a blinded manner by several experimenters with the identity of mice and experimental day unknown during AIS measurements. Only AIS whose entire length was fully contained in a single optical plane (that is, individual z-stack frame) were tracked for further analysis. For calcium imaging experiments, Suite2p was used to extract calcium traces of individual neurons. False positives were discarded and cells were defined manually if these were not detected automatically. Neuropil signal around each cell was multiplied by 0.7 and subtracted from the calcium trace with the result considered as the calcium signal of a single neuron. For ex vivo imaging of c-Fos expression and AIS length, images were acquired with an LSM900 confocal

microscope (Zeiss). Confocal images were postprocessed with ImageJ (Version 2.0). For quantification and thresholding of c-Fos expression, the ImageJ plugin Quanty-cFOS⁶⁰ was used, and the SNT⁶¹ plugin was utilized to track AIS.

For in vivo–ex vivo alignment experiments, matching between two-photon fields of view and ex vivo horizontal mPFC images was achieved by performing a 2D z-projection in ImageJ of manually selected confocal planes. The planes were manually selected by identifying topological hallmarks present in both the two-photon and the confocal images.

Finally, for super-resolution imaging, a STED microscopy setup was used (Facility Line, Abberior Instruments). The setup is built on a motorized inverse microscope body (IX83, Olympus) and equipped with a pulsed 775-nm depletion laser and 405-nm, 485-nm, 561-nm and 640-nm excitation lasers. Spectral detection was performed using single-photon-counting avalanche photodiode detectors. Acquisition was normalized to 5–15 photons per pixel and μ s with a pixel dwell time of 5 μ s. Channel-specific line accumulation was individually adapted to reach a photon count above 100 within the brightest pixels per field of view. Spectral windows were set to 410–470 nm for Alexa405Plus, 510–600 nm for 460 L, 600–630 nm for StarOrange and 650–725 nm for StarRed fluorophors. Images were acquired using a $\times 60$ oil immersion objective (1.45 NA, UPLXAP060XO, Olympus) and a set lateral pixel size of 20 nm.

Image analysis of AIS recorded via GRIN-lens-based two-photon imaging

Due to the internal magnification of GRIN lenses and optical nonuniformities^{62,63}, in vivo AIS lengths are provided as relative changes to themselves, providing an accurate way of comparing AIS across different planes and samples. The start and the end of the AIS were defined visually by the experimenter in two-photon in vivo images. To compare changes in AIS length relative to the first imaging session (habituation, Fig. 2d) the mean of AIS length during the habituation session was calculated per mouse. Next this value was subtracted from each individual AIS length per mouse and the result divided by the same mean value using equation (1) as follows:

$$\text{Normalized AIS length}_i = \frac{(\text{AIS length}_i - \text{mean}(\text{AIS length})_{\text{hab}})}{\text{mean}(\text{AIS length})_{\text{hab}}} \times 100 \quad (1)$$

To calculate the mean change in AIS length within steady-state sessions versus the change in length within the learning phase for each plasticity cluster (Fig. 3c), the normalized AIS length (the relative length change to habituation session) was summed across steady-state sessions and across the learning phase of each plasticity cluster per individual AIS (i) and divided by the number of sessions according to equation (2):

$$\text{Session change}_i = \frac{(\text{norm AIS length}_{n(i)} + \text{norm AIS length}_{n(i)+1})}{2} \quad (2)$$

where for the learning phase n equals Extinction session 1 and $n + 1$ equals Extinction session 2 and for the steady-state phase n equals Steady day 1 and $n + 1$ equals Steady day 2. The resulting session change values were averaged per cluster.

Histology

Following completion of behavioral experiments, mice were transcardially perfused with around 5 ml phosphate buffered saline (PBS, ThermoFisher) followed by 20 ml 4% paraformaldehyde (PFA) in PBS (pH = 7.4). Brains were removed and stored overnight in 4% PFA. Coronal slices (150 μ m) were prepared using a vibratome (Leica Biosystems VT1000S) and immunostained for AnkG and Na_v1.6 using the following solutions and protocol: carrier solution: 1% normal horse

serum (Vector Laboratories) with 0.5% Triton (ThermoFisher) in PBS (ThermoFisher), blocking solution: 10% normal horse serum with 0.5% Triton in PBS, PBST solution: 0.1% Triton in PBS. After 3 rounds of PBS washes, slices were blocked for 2 h at room temperature and incubated in primary antibodies in carrier solution (guinea pig anti-AnkG, 1:500, Synaptic Systems, cat. no. 386005; rabbit anti-Na_v1.6, 1:150, Alomone, cat. no. ASC-009) overnight at 4 °C. Slices were washed again 3 times in PBST and incubated for 2 h at room temperature in secondary antibodies in carrier solution (goat DyLight405 anti-guinea pig, 1:400, Jackson ImmunoResearch, cat. no. 160-475-003; goat AlexaFluor467 anti-rabbit, 1:400, Invitrogen, cat. no. A21244). After 3 final washes, slices were mounted on slides and cover slipped using 22 $\times$ 50 mm, 0.16–0.19-mm-thick cover glass (FisherScientific). Images were acquired with an LSM700 upright confocal microscope (Zeiss). Confocal images were postprocessed with ImageJ (Version 2.0). The confocal tile scan overview image in Fig. 4d was adjusted for differences in contrast and brightness of different planes at tile edges in a plane-by-plane manner. This adjustment was not required for the quantification of individual AIS at high magnification.

For c-Fos experiments, timed perfusions were carried out 90 min after the start of the behavior for each animal, which included a 20-min home cage period after the end of the behavioral session. Brain tissue was kept in 4% PFA. Coronal sections of mPFC of 100 μ m were prepared with a Leica VT1000S vibratome. Antibody staining protocols were the same as described. Antibodies: c-Fos: rat anti-c-Fos, 1:1000, Synaptic Systems, cat. no. 226017; secondary antibody: Alexa Fluor Plus 555 donkey anti-rat, 1:1000, Invitrogen, cat. no. A48270. AnkG was co-stained as described.

For the experiments where in vivo neural activity was matched to ex vivo AIS length, tissue sectioning was performed in the horizontal plane (200 μ m thickness). The immunostaining procedure to label AnkG was the same as described.

For 2D-STED microscopy, tissue after perfusion was embedded in OCT compound (Tissue Tek, Sakura Finetek) and lowered into –160 °C isopentane for quick, unidirectional freezing. Coronal slices (40 μ m) were prepared using a cryostat (Leica Biosystems CM3050 S) and transferred directly into 2% PFA for 10 min. After washing in PBS, slices were incubated in blocking buffer (1% BSA, 0.2% fish skin gelatine, 0.1% Triton in 0.1 M PBS) for 60 min and subsequently incubated in primary antibodies (guinea pig anti-AnkG, 1:1,000, Synaptic Systems, cat. no. 386005; rabbit anti-Na_v1.6, 1:500, Alomone, cat. no. ASC-009; rabbit anti-c-Fos, 1:500, Cell Signaling, cat. no. 2250; chicken anti-NeuN, 1:1,000, cat. no. 266006, Synaptic Systems) overnight at 4 °C. After 3 rounds of PBS washes, slices were incubated for 2 h in secondary antibodies (Alexa 405 Plus goat anti-chicken, 1:1000, Invitrogen, cat. no. A48254; Star 460 L goat anti-rabbit, 1:1,000, Abberior, cat. no. ST460L-1002; StarOrange goat anti-mouse, 1:1,000, Abberior, cat. no. STORANGE-1001; StarRed goat anti-guinea pig, 1:1,000, Abberior, cat. no. STRED-1006) at room temperature. Slices were washed again 3 times in PBS, mounted in an aqueous embedding medium (Abberior Mount Solid Antifade, Abberior GmbH) and cover slipped using precision cover glasses with a thickness of 0.170 mm $\pm$ 0.05 mm (Paul Marienfeld GmbH).

Statistical analysis

Statistical analysis was performed using Matlab (Mathworks) and Python (Version 3.8, Python Software Foundation including SciPy and scikit-learn). Normality was tested using the D'Agostino and Pearson's test. To identify groups of AIS with distinct length dynamics, k -means clustering using the k -means+ plus initialization method (Python and scikit-learn) and a restriction to four clusters was applied. Additional attempts at increasing the number of clusters resulted in the formation of subclusters that were still part of the four primary ones. For analysis of AIS length comparing c-Fos⁺ and c-Fos[−] cells, Kolmogorov–Smirnov test and CDF analysis were used. Values are presented as mean $\pm$ s.e.m. unless stated otherwise. Box and whisker plots indicate mean (dot) and

median (horizontal line), interquartile range as well as the minimum to maximum value of the distribution. No statistical methods were used to predetermine sample sizes but our sample sizes are similar to those reported in previous publications in the field. Animals were randomly added to experimental groups without preselection. All animals with visible AIS were included in the experiments and analysis, except one mouse which only had two visible AISs. Micrographs in Fig. 1a could be repeated in $N = 10$ mice. Micrographs in Fig. 1b could be repeated for $n = 5$ AIS from $N = 1$ mouse (Extended Data Fig. 1. The overlap of AnkG and Na_v channels was established previously³⁴). Statistical tests are given in the text or figure legends. *, ** and *** indicate P values smaller than 0.05, 0.01 and 0.001, respectively.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Data for code are available via Zenodo at <https://doi.org/10.5281/zenodo.17405776> (ref. 64). Source data are provided with this paper.

Code availability

Analysis code is available via Zenodo at <https://doi.org/10.5281/zenodo.17405746> (ref. 65).

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Acknowledgements

We thank A. Karpova and D. Schaffer, who gifted the rAAV2-retro helper plasmid (Addgene plasmid 81070V). We thank R. Strittmatter, R. Häring and S. Saner for workshop and electrical engineering support and J. A. Taylor, J. Amorim Freire and M. Hasegawa for

experimental advice. V. Bennett and P. Jenkins originally generated the AnkG-GFP mouse line. The STED microscope is part of the Imaging Core Facility at Johannes Kepler University, Medical Faculty, Linz, Austria. This work was supported by the Swiss National Science Foundation (SNF Professorship PP00P3_170672 to J.G., Project Grant 189123 to A.L.), the European Research Council (Starting Grant 803870 to J.G.), the Forschungspreis der Schweizerischen Hirnliga (A.L. and J.G.), the Deutsche Forschungsgemeinschaft (Walter Benjamin Program PN 458054460 to C.T. and R.P.-M.; DFG EN 1240/2-1 to M.E.; SFB 1089 to J.G. and S.K.; SPP2411 to J.G.), Horizon 2020 ICT (DEEPER 101016787 and the IIT Brain and Machines Flagship Program to T.F.), the iBehave program 'Netzwerke 2021', an initiative of the Ministry of Culture and Science of the State of North Rhine-Westphalia (J.G. and S.K.), Dementia Research Switzerland—Synapsis Foundation (2018-CDA02 and 2020-CDA04 to S.K.), the Department of Biomedicine at the University of Basel, Switzerland (J.G.), the Research Fund of the University of Basel, Switzerland (D.A.G.), the IMPETUS program at Johannes Kepler University (J.M.J.) and the Deutsches Zentrum für Neurodegenerative Erkrankungen (DZNE), Bonn, Germany (J.G. and S.K.). The funders had no role in study design, data collection and analysis, decision to publish or preparation of the manuscript.

Author contributions

Conceptualization: C.M.B., D.A.G., S.K., A.L., J.G. Methodology: C.M.B., D.A.G., C.T., M.E., A.S., T.F., J.G. Investigation two-photon imaging: C.M.B., D.A.G., R.P.-M. Investigation histology: C.M.B., R.P.-M., J.E., J.M.J., M.E., S.M.I. Data curation: C.M.B., D.A.G., R.P.-M., J.E., J.M.J., M.E. Formal analysis two-photon imaging: C.M.B., D.A.G., R.P.-M. Formal analysis histology: C.M.B., R.P.-M., J.E., J.M.J., M.E. Visualization: C.M.B., D.A.G., R.P.-M., J.E., J.M.J., M.E., J.G. Funding acquisition: J.G., A.L., M.E., T.F., S.K., R.P.-M., D.A.G. Supervision: J.G. Writing—original draft: D.A.G., C.M.B., J.G. Writing—review and editing: all authors.

Funding

Open access funding provided by Deutsches Zentrum für Neurodegenerative Erkrankungen e.V. (DZNE) in der Helmholtz-Gemeinschaft.

Competing interests

The authors declare no competing interests.

Additional information

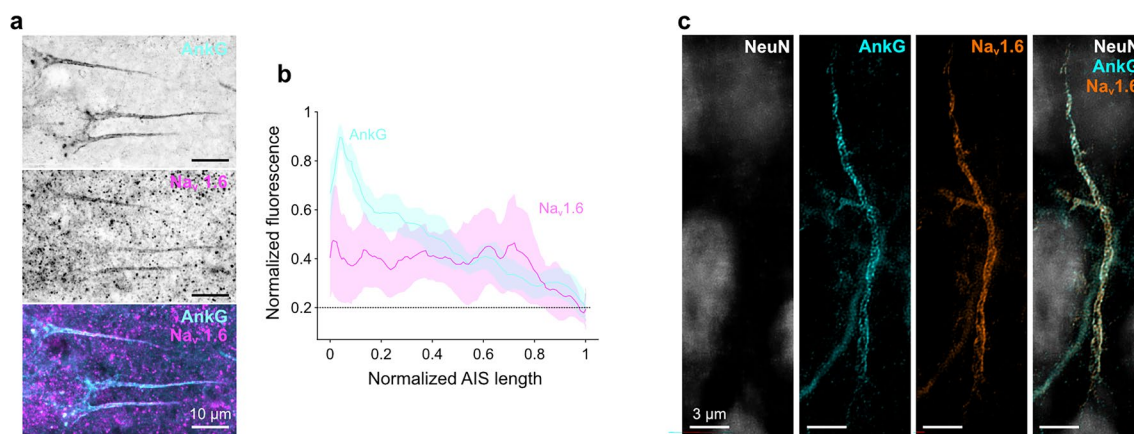
Extended data is available for this paper at <https://doi.org/10.1038/s41593-025-02152-5>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41593-025-02152-5>.

Correspondence and requests for materials should be addressed to Dan Alin Ganea or Jan Gründemann.

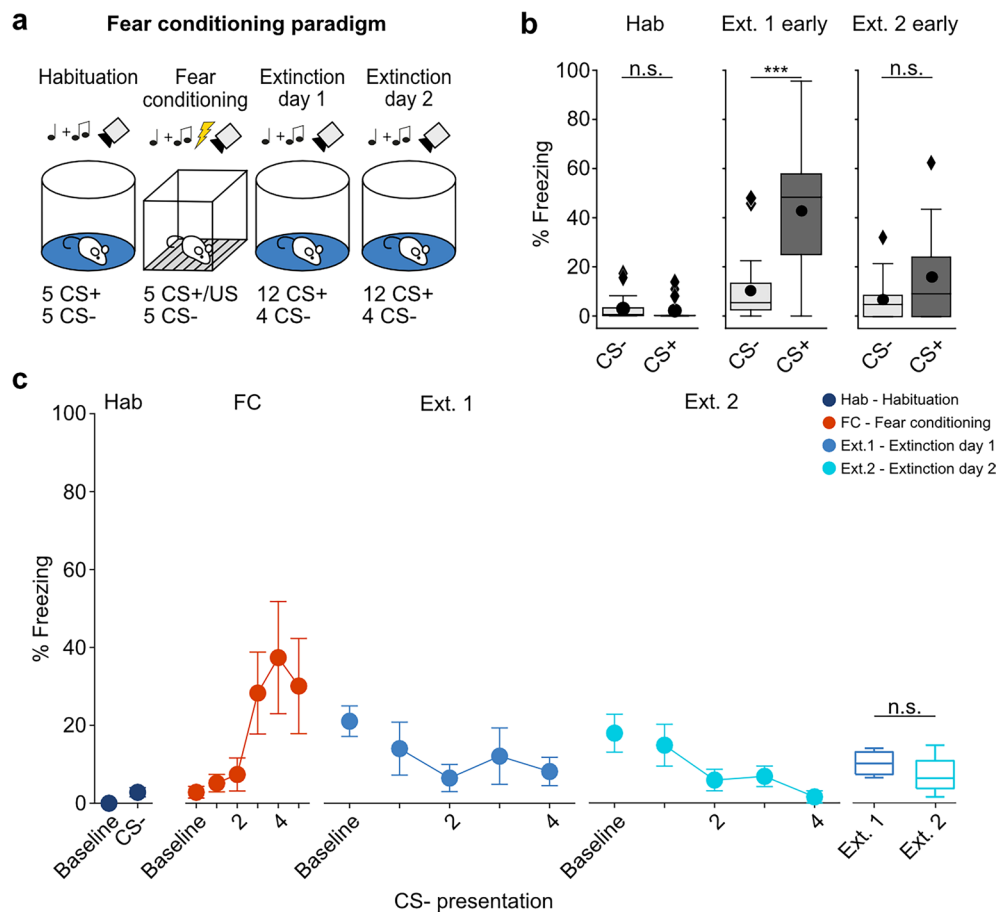
Peer review information *Nature Neuroscience* thanks Amélie Freal and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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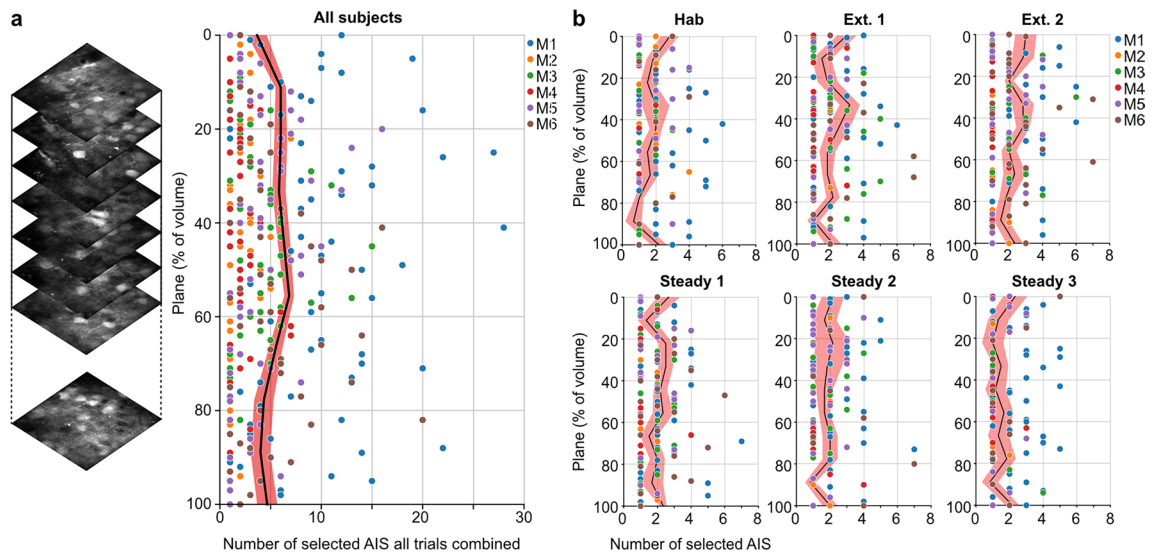
Extended Data Fig. 1 | AnkG immunofluorescence overlaps with Nav1.6 in mPFC pyramidal neurons. **a**, Confocal image of layer II/III pyramidal neurons in mPFC co-stained against AnkG (cyan) and Nav1.6 (magenta). Inverted black & white image of AnkG (top), Nav1.6 (middle) and color-coded merge image. **b**, Normalized fluorescence intensity profile of AnkG and Nav1.6 detected along

mPFC pyramidal neurons AIS (n = 5 AIS from N = 1 mouse). Data are presented as mean values $\pm$ SEM. **c**, Immunofluorescence and STED imaging of NeuN, AnkG and Nav1.6. Merge image is the same as Fig. 1b. STED imaging was reproduced for N = 3 mice.



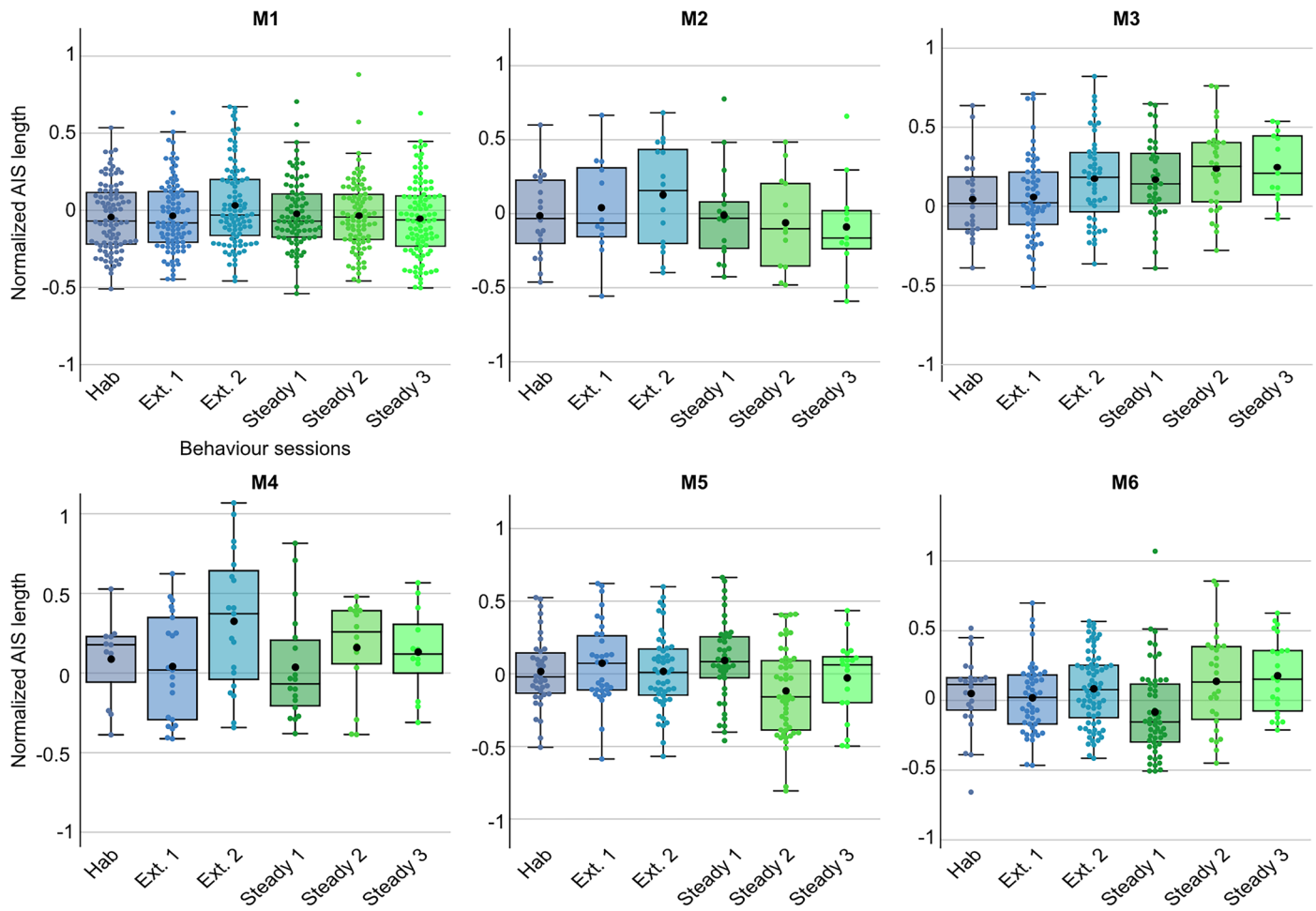
Extended Data Fig. 2 | Behavioral paradigm and freezing to the CS during auditory fear conditioning. **a**, Schematic of the auditory fear conditioning paradigm. **b**, Freezing to the CS+ and CS- during auditory fear conditioning. (Hab CS+ vs CS-: $P = 0.24$, Ext. 1 early CS+ vs CS-: $P = 1.12 \times 10^{-5}$ (CS+ $42.6 \pm 2.5\%$, CS- $10.2 \pm 1.3\%$), Ext. 2 early CS+ vs CS-: $P = 0.079$ (CS+ $16.7 \pm 3.7\%$, CS- $7.3 \pm 1.9\%$); two-sided Mann Whitney U test; $N = 6$ mice). **c**, Neutral stimulus (CS-) freezing

(mean $\pm$ s.e.m.) during the habituation, fear conditioning as well as extinction days (Ext. 1, Ext. 2., CS- Ext. 1 vs Ext. 2: $P = 0.46$; two-sided Mann Whitney U test, $N = 6$ mice). Boxplots represent median, 2nd, 3rd quartile, minimum and maximum. Dot indicates mean and diamond outliers. *** indicate p-values smaller than 0.001.



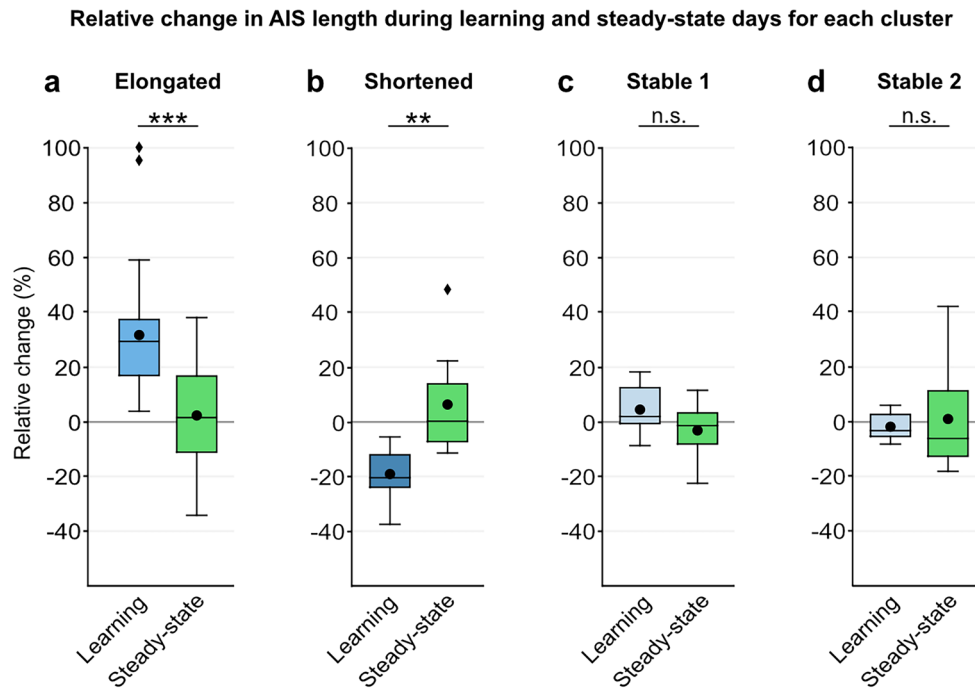
Extended Data Fig. 3 | Number of selected AIS per plane for the different imaging sessions. a, Overall number of AIS tracked per imaging plane for all subjects (5.03 ± 0.26 AIS selected per plane) all sessions combined. Black and red lines represent mean $\pm$ s.e.m. **b**, Overall number of AIS tracked per imaging plane

for all subjects (2.06 ± 0.12 AIS selected per plane) during the different imaging days. Mean $\pm$ s.e.m for Hab: 2.06 ± 0.12 AIS selected per plane, Ext. 1: 2.09 ± 0.12 , Ext. 2: 2.12 ± 0.12 , Steady 1: 1.98 ± 0.11 , Steady 2: 2.06 ± 0.13 , Steady 3: 1.98 ± 0.13 .



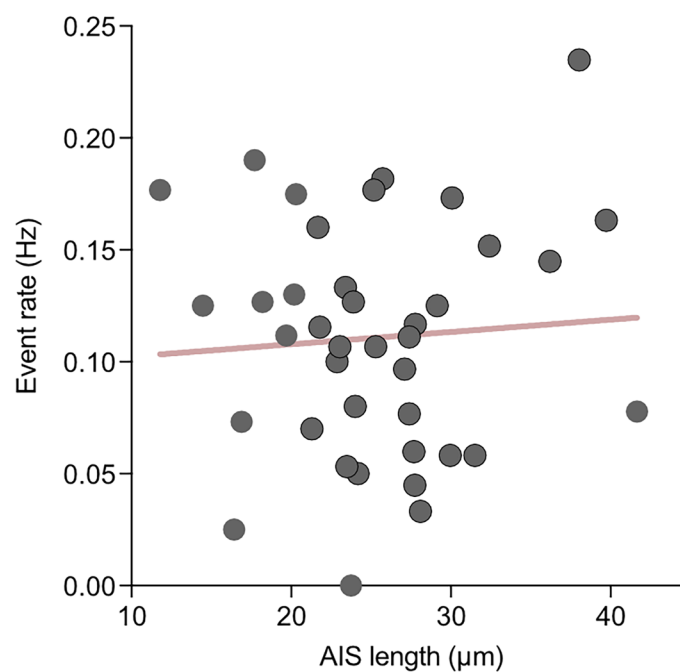
Extended Data Fig. 4 | AIS length population change for each subject. Relative change in AIS length normalized to the population mean of all animals pre-learning (%) for each subject (M1: $n = 645$ AIS, M2: $n = 102$ AIS, M3: $n = 236$ AIS, M4:

$n = 97$ AIS, M5: $n = 257$ AIS, M6: $n = 268$ AIS). Boxplots represent median, 2nd, 3rd quartile, minimum and maximum. Black dots indicate means and colored dots individual data points.

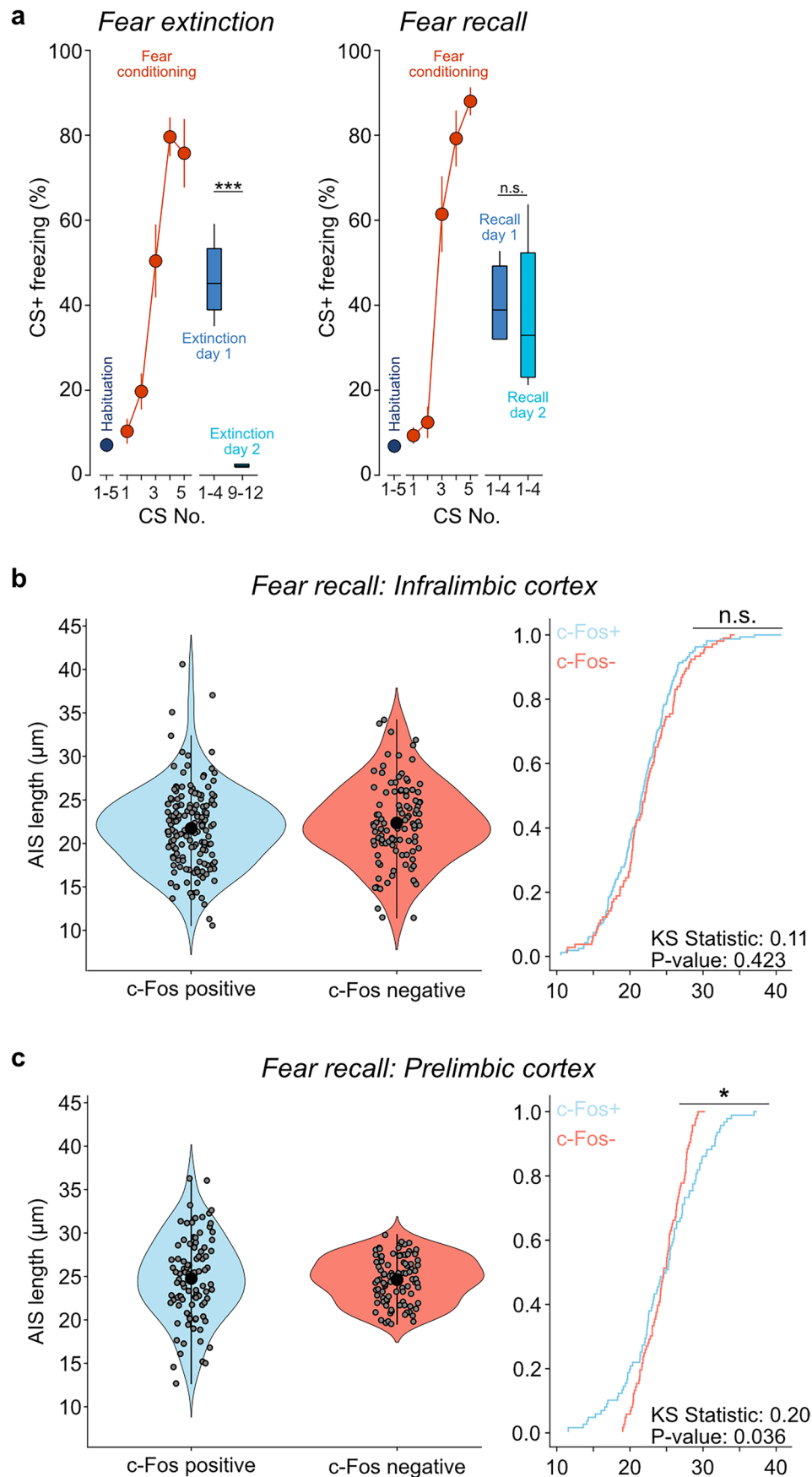


Extended Data Fig. 5 | AIS plasticity during learning and the steady-state days for each cluster. Total percent change in length within learning and steady-state days for each cluster (**a**, Elongated cluster: learning vs steady-states, $P = 1.77 \times 10^{-5}$; **b**, Shortened cluster: learning vs steady-states, $P = 1.71 \times 10^{-3}$; **c**, Stable 1 cluster: learning vs steady-states, $P = 0.11$; **d**, Stable 2 cluster: learning vs steady-states, $P = 0.74$; two-sided Mann Whitney U test, $N = 6$ mice). Learning and Steady-state

respectively represent the summation of the total change during the learning days (Ext.1 and Ext.2) and Steady-state days (steady-state day 1 and steady-state day 2). Boxplots represent median, 2nd, 3rd quartile, minimum and maximum. Dot indicates mean and diamond outliers. **, *** indicate P -values smaller than 0.01 and 0.001, respectively.



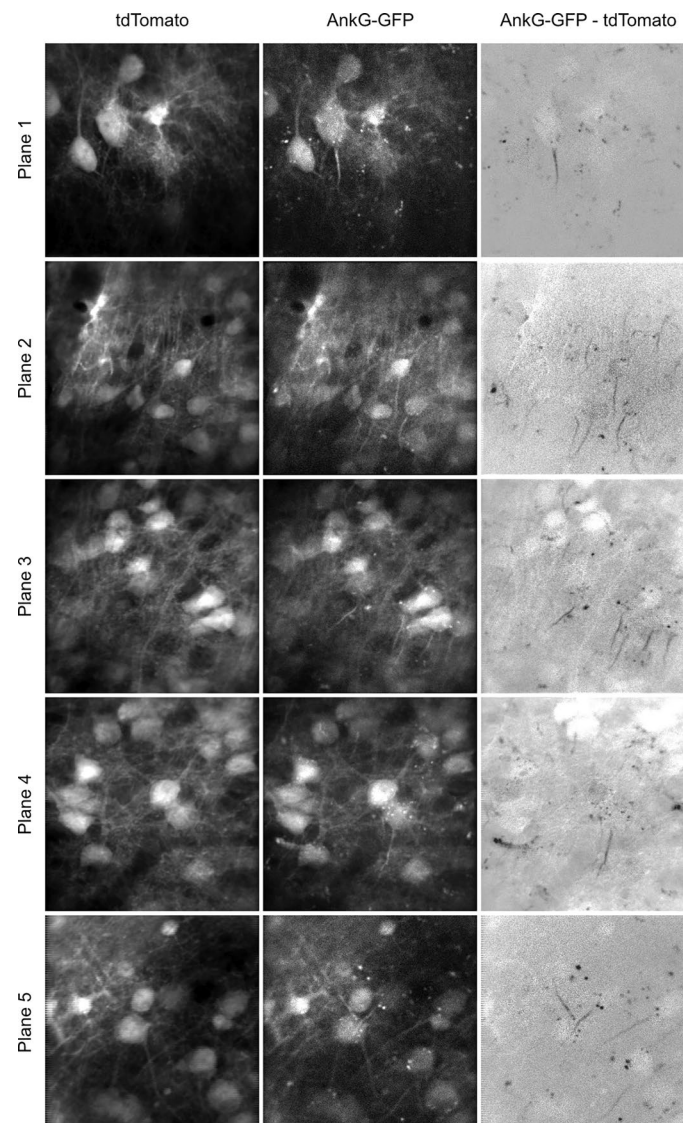
Extended Data Fig. 6 | AIS length is not correlated with in vivo event rate. Scatter plot of correlation between in vivo event rate and ex vivo measured AIS length of cells that could be identified in in vivo Ca^{2+} imaging and post hoc ex vivo confocal imaging. Slope = 0.0005, $r^2 = 0.005$, $P = 0.6769$. Pearson correlation; $n = 38$ aligned neurons from 3 mice.



Extended Data Fig. 7 | See next page for caption.

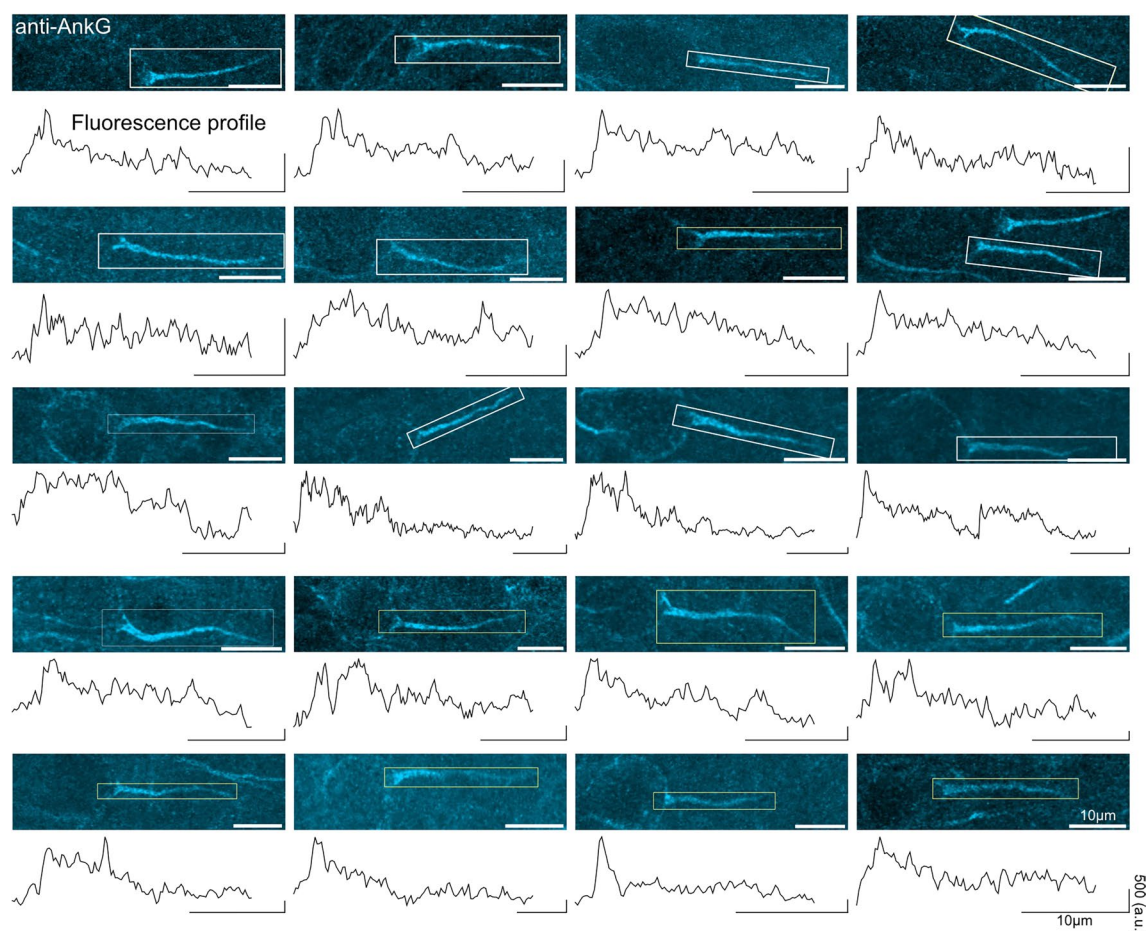
Extended Data Fig. 7 | AIS length and c-FOS expression in mPFC neurons upon fear recall. **a**, Freezing behaviour of mice during the fear extinction (left, $p = 1.59 \times 10^{-12}$, $N = 9$ mice; Mann Whitney U test) and fear recall paradigms (right, $p = 0.795$, $N = 9$ mice; two-sided Mann Whitney U test). Boxplots represent median, 2nd, 3rd quartile, minimum and maximum. **b**, Violin plots of AIS length for c-Fos positive and negative cells in infralimbic cortex after fear recall (left, c-Fos+ mean = $21.74 \pm 0.37 \mu\text{m}$, c-Fos- mean = $22.35 \pm 0.45 \mu\text{m}$). Population cumulative distribution function (CDF, right, $p = 0.423$ and KS statistic = 0.11,

$n = 268$, $N = 9$ mice; two-sided Kolmogorov-Smirnov test). **c**, Violin plots of AIS length for c-Fos positive and negative cells in prelimbic cortex after fear recall (left, c-Fos+ mean = $24.19 \pm 0.49 \mu\text{m}$, c-Fos- mean = $24.06 \pm 0.27 \mu\text{m}$). Population CDF (right, $p = 0.036$ and KS statistic = 0.20, $n = 189$, $N = 7$ mice; two-sided Kolmogorov-Smirnov test). Violon plots represent 2nd, 3rd quartile, minimum and maximum. Black dots indicate mean. *, *** indicate P-values smaller than 0.05 and 0.001, respectively.



Extended Data Fig. 8 | In vivo two-photon imaging of axon initial segments of mPFC neurons. Representative in vivo two-photon images of mPFC neurons co-expressing AnkG-GFP and tdTomato from five imaging planes. To emulate

a background subtraction, tdTomato images were subtracted from AnkG-GFP images revealing higher expression of AnkG-GFP at the AIS (results observed in N = 6 mice).



Extended Data Fig. 9 | Ex vivo immunofluorescence of Ankyrin-G and AIS fluorescence profile. Confocal images of Ankyrin-G immunofluorescence in mPFC neurons. Axon initial segments (AIS) typically start at the soma in mPFC, indicated by a sharp increase in Ankyrin-G fluorescence intensity. Please note the AnkG immunofluorescence stain of the somatic membrane. Traces indicate

fluorescence intensity profiles along the length of the white boxes. AIS starts can be clearly identified as a sharp rise in fluorescence visually in the images as well as in the fluorescence intensity traces. AIS typically start directly at the soma. Results observed in N = 18 mice.

Corresponding author(s): Jan Gründemann

Last updated by author(s): Oct 21, 2025

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Software and code

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Data collection	Independent Neuroscience Services two-photon microscope equipped with Ti:sapphire laser (Chameleon Vision S, Coherent), resonant scanning system (CRS08K, Cambridge Technology), photomultiplier tubes (PMT2102, Thorlabs), ScanImage software (Versin 5.7, Vidrio. Technologies), LSM900 confocal microscope (Zeiss) and a stimulated emission depletion (STED) microscope (Facility Line, Abberior Instruments) built on a motorized inverted microscope body (IX83, Olympus) were used to acquire images. Behavior experiments were performed using Tucker Davis Technologies, RZ6 and Ethovision XT 14 (Noldus).
Data analysis	Data were motion corrected and analysis using MatLab (Mathworks) , Python (Version 3.8, Python Software Foundation incl. SciPy and scikit-learn) and ImageJ Version 2.0 (Simple Neurite Tracer, SNT and Quanty-cFos). The analysis code is available at Zenodo, DOI: 10.5281/zenodo.17405746

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size Sample sizes experiments were based on previously established publications. No statistical methods were used.

Data exclusions Provided under Materials and Methods, Statistical analysis.

Replication All imaging sessions were performed prior to analysis according to Materials and Methods, Behavioral paradigm.

Randomization Provided under Materials and Methods, Statistical analysis.

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☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

Provided in Materials and Methods, Surgical procedures and Histology. Primary antibodies: Guinea-pig anti-ankyrin-G, 1:500, Synaptic systems, Cat# 386005; rabbit anti-Nav1.6, 1:150, Alomone, Cat# ASC-009; rat anti c-fos, 1:1000, Synaptic Systems, Cat#226017; guinea-pig anti-ankyrin-G, 1:1000, Synaptic Systems, Cat# 386005; rabbit anti-Nav1.6, 1:500, Alomone, Cat# ASC-009, rabbit anti-c-Fos, 1:500, Cell Signaling, Cat# 2250; chicken anti-NeuN, 1:1000, Cat# 266006, Synaptic Systems and secondary antibodies: goat DyLight405 anti-guinea-pig, 1:400, Jackson Immuno Research, Cat# 160-475-003; goat AlexaFluor467 anti-rabbit, 1:400, Invitrogen, Cat# A21244; alexaFluor Plus 555 donkey anti-rat, 1:1000, Invitrogen, Cat#A48270; Alexa 405 Plus goat anti-chicken, 1:1000, Invitrogen, Cat# A48254; Star 460L goat anti-rabbit, 1:1000, Abberior, Cat# ST460L-1002; StarOrange goat anti-mouse, 1:1000, Abberior, Cat# STORANGE-1001; StarRed goat anti-guinea-pig, 1:1000, Abberior, Cat# STRED-1006.

Validation

Commercial antibodies were validated by manufacturers.

Animals and other research organisms

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Laboratory animals

Provided in Materials and Methods, Animal subjects. Experiments were conducted in 9-15 weeks old male, C57Bl/6J, Ank-G-GFP or Ank-G-GFP::Ai9 mice. Cre-dependent Ank-G-GFP mice (B6;129SVank3tm1DCI/ HD) were provided by Maren Engelhardt, Johannes Kepler University Linz, Austria. Ank-G-GFP mice were crossed with cre-dependent tdTomato reporter (Ai9) mice (Gt(ROSA)26Sortm9(CAG-tdTomato)Hze, JAX: IMSR_JAX:007909 (55)) to generate the Ank-G-GFP/Ai9 mice. Mice were housed on a 12-hour light / dark cycle and had ad libitum access to food and water.

Wild animals

The study did not involve wild animals.

Reporting on sex

Only male mice were used in this study

Field-collected samples

The study did not involve animals collected from the field.

Ethics oversight

Experiments were performed in accordance with the institutional guidelines of the University of Basel (University of Basel, Tierschutz) and the DZNE Bonn were approved by the Cantonal Veterinary Office of Basel-Stadt, Switzerland and the LANUV, NRW, Germany.

Note that full information on the approval of the study protocol must also be provided in the manuscript.